

Ash Rauch

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Cybersecurity Student | Aspiring Penetration Tester

Computer Science senior specializing in cybersecurity with strong foundations in Linux systems, networking, and security analysis. Experienced in analyzing program behavior with a focus on penetration testing and vulnerability assessment. Seeking opportunities focused on security analysis and offensive security.

SKILLS

Security & Systems

Kali Linux • Unix • Networking Fundamentals • Virtual Machines • Vulnerability Analysis
Security Assessment • Security Analysis • Offensive Security Fundamentals

Technical Tools

Git/Github • Bash • VScode • Virtualization • Python • Java • C • HTML/CSS

Soft Skills

Teamwork • Leadership • Customer Service • Time Management

PROJECT EXPERIENCE

Cyber Defense Security Labs (Kali Linux/Ubuntu), JMU

Jan 26 - Present

- Perform hands-on offensive security and penetration testing exercises in virtualized Linux environments.
- Conduct vulnerability discovery, exploitation, and system analysis using industry-standard security tools.
- Apply concepts across authentication, memory vulnerabilities, web app security, and privilege escalation.
- Expand practical penetration testing skills through structured cybersecurity lab environments.

Mini-ELF Interpreter Analysis (C/VSCode), JMU

Nov 25

- Analyzed executable instruction behavior and system execution flow.
- Investigated register operations and memory interactions through structured debugging.
- Developed low-level program execution understanding applicable to vulnerability analysis.

UnitED Unit Conversion Calculator (Java/Eclipse), JMU

Mar 24 - May 24

- Collaborated within a five-person development team to design and deliver a functional GUI application.
- Contributed interface components and usability improvements through testing and iteration.

Origins+, Minecraft Mod Addon (Java/IntelliJ), Personal Project

Aug 21 - Dec 21

- Modified and extended functionality within an existing software environment.
- Applied troubleshooting and problem-solving techniques to maintain stability and usability.
- Demonstrated independent technical learning and project ownership.

EXPERIENCE

Web Development Intern (HTML, CSS, Elfsight), Let's Bouncz Inflatables

Oct 24 - Nov 25

Harrisonburg, VA

- Redesigned and expanded company website to improve usability and customer booking workflows.
- Implemented navigation and interface improvements enhancing customer interaction.
- Collaborated with business leadership to translate operational needs into technical solutions.

Driver/Supervisor, Let's Bouncz Inflatables

Apr 24 - Nov 25

Harrisonburg, VA

- Coordinated delivery logistics and supervised team operations in fast-paced client environments.
- Developed leadership, communication, and operational decision-making skills.

Other Roles

Lifeguard II, Oak Marr Rec Center | Barista, Kung Fu Tea | Barista, Starbucks

May 19 - Aug 23

- Built transferable skills in teamwork, customer service, and working under pressure.

EDUCATION

James Madison University (JMU), Harrisonburg, VA

May 26

Bachelor of Science in Computer Science

NSA Cybersecurity Certificate (upon graduation)

CAMPUS AND COMMUNITY

Member, Outdoor Adventure Club, Harrisonburg, VA

Aug 23 - Present

Eagle Scout, Scouts BSA, Vienna, Virginia

Feb 20